

REQUIREMENTS:

1. Authentication Module

- Email Login
- Google Login
- OTP Verification
- Forgot Password
- JWT Authentication
- Role-Based Access Control

2. Course Management

- Create Courses
- Edit Courses
- Delete Courses
- Course Categories
- Learning Paths
- Course Progress
- Prerequisites
- Recommended Courses

3. Learning Module

Supports:

- PDF Notes
- PPT
- Videos
- Interactive Slides
- Coding Labs
- Flashcards
- Practice Exercises

Features

- Continue Learning

- Resume Progress
- Bookmark Topics
- Learning History

4. AI Tutor Module

Features

- Explain Concepts
- Answer Student Doubts
- Generate Examples
- Explain Code
- Summarize Topics

5. Practice Module

Practice Types

- MCQs
- Coding Questions
- Fill in the Blanks
- True/False
- Match the Following
- Subjective Questions

Features

- Adaptive Difficulty
- Unlimited Practice
- Instant Feedback

6. Assessment Module

Supports

- Quiz
- Mid Exam
- Final Exam
- Coding Contest

- Timed Assessment

Features

- Random Questions
- Auto Submission
- Negative Marking
- Time Limit
- Section-wise Assessment
- Auto Evaluation
- Manual Evaluation

7. Secure Examination Module

- Tab Switching Detection

8. Assignment Module

- Faculty
- Create Assignment
- Deadline
- Rubrics
- Attach Files
- Student
- Submit Assignment
- View Feedback

11. Analytics Dashboard

Student Dashboard

- Learning Progress
- Course Completion
- Assessment Scores
- Practice Time
- Weak Topics
- Streak

- XP

Faculty Dashboard

- Student Performance
- Course Analytics
- Question Difficulty Analysis
- Completion Statistics

Admin Dashboard

- Active Users
- Courses
- Revenue (if applicable)
- Assessments
- Pass Percentage
- System Usage

12. Certification Module

Features

- Auto Certificate Generation
- PDF Download
- QR Code Verification
- Digital Signature
- Certificate Verification Portal

13. Gamification Module

Features

- XP Points
- Levels
- Daily Streak
- Weekly Challenge
- Monthly Challenge

- Leaderboard
 - Badges
 - Achievement System
-

14. Notification Module

Notifications

- Email
- SMS (Optional)
- In-App Notifications
- Push Notifications

Alerts

- Assignment Due
 - Quiz Reminder
 - Course Updates
 - Certificate Ready
 - New Course
 - Assessment Results
-

15. Reports Module

Generate Reports

- Student Report
- Faculty Report
- Course Report
- Assessment Report
- Question Analysis
- Completion Report
- Attendance Report
- Learning Analytics

NON FUNCTIONAL REQUIREMENTS:

Responsive Web Application

Mobile-Friendly Interface

High Availability

Scalable Architecture

Secure Authentication

Fast Response Time (<2 seconds for most operations)

Cloud Deployment Support

Database Backup & Recovery

RESTful API Design

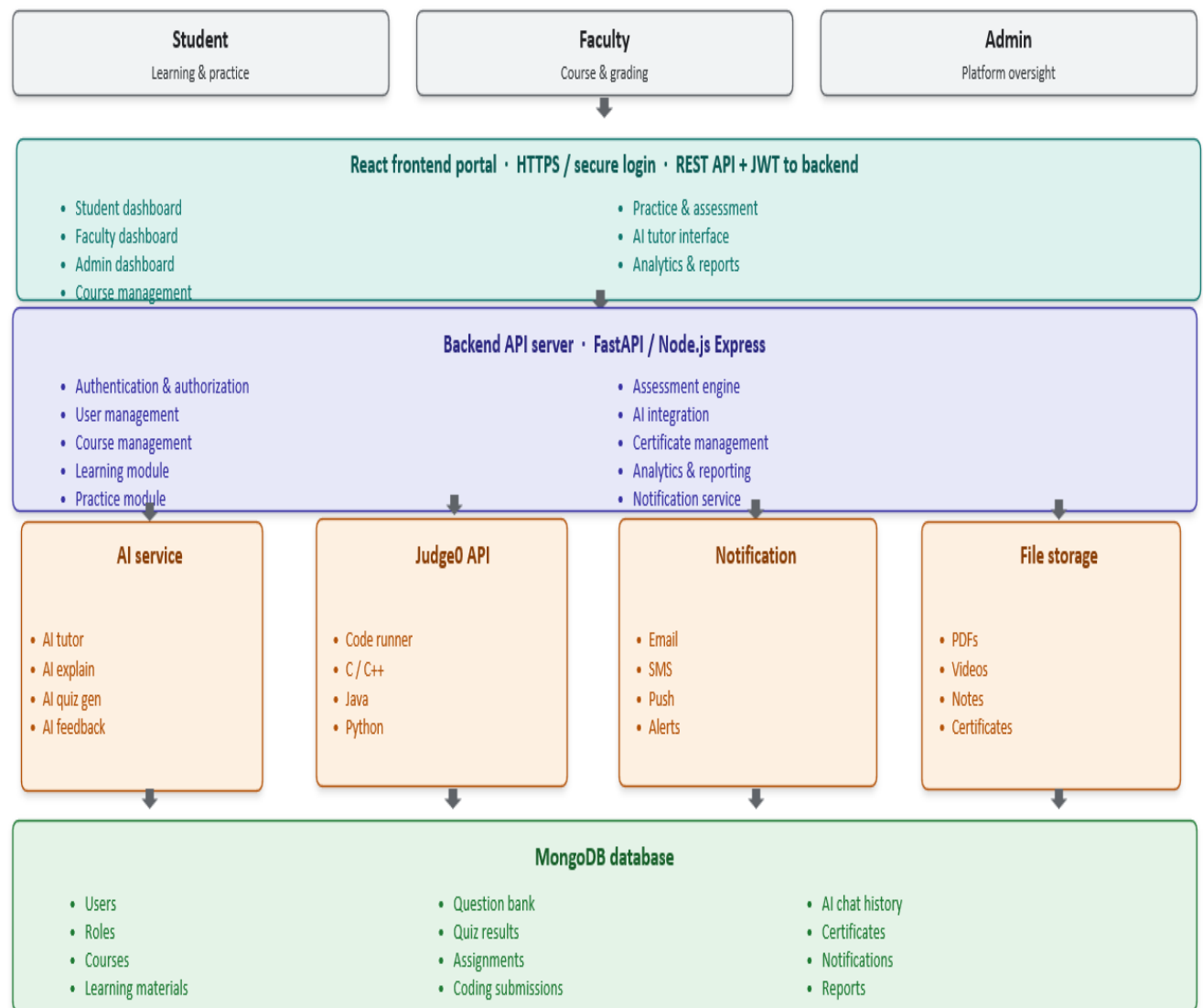
Docker Support

Easy Maintenance

Modular Architecture

System Architecture – Learning Practice and Assessment Portal:

LEARNING AND ASSESSMENT platform — system architecture



Database Schema

1. ROLE

CREATE TABLE ROLE (

Role_ID INT PRIMARY KEY AUTO_INCREMENT,

Role_Name VARCHAR(50) NOT NULL UNIQUE,

Description VARCHAR(255)

);

2. USER

```
CREATE TABLE USER (  
    User_ID    INT PRIMARY KEY AUTO_INCREMENT,  
    Name       VARCHAR(100) NOT NULL,  
    Email      VARCHAR(150) NOT NULL UNIQUE,  
    Password   VARCHAR(255) NOT NULL,  
    Phone      VARCHAR(20),  
    Profile_Image VARCHAR(255),  
    Role_ID    INT NOT NULL,  
    Status     VARCHAR(20) DEFAULT 'Active',  
    Created_At TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (Role_ID) REFERENCES ROLE(Role_ID)  
        ON UPDATE CASCADE ON DELETE RESTRICT  
);
```

3. COURSE

```
CREATE TABLE COURSE (  
    Course_ID  INT PRIMARY KEY AUTO_INCREMENT,  
    Faculty_ID INT NOT NULL,  
    Title      VARCHAR(150) NOT NULL,  
    Description TEXT,  
    Category   VARCHAR(100),  
    Duration   VARCHAR(50),  
    FOREIGN KEY (Faculty_ID) REFERENCES USER(User_ID)  
        ON UPDATE CASCADE ON DELETE RESTRICT  
);
```

4. LEARNING_MODULE


```

CREATE TABLE LEARNING_MODULE (
    Module_ID INT PRIMARY KEY AUTO_INCREMENT,
    Course_ID INT NOT NULL,
    Module_Name VARCHAR(150) NOT NULL,
    Video_URL VARCHAR(255),
    PDF_URL VARCHAR(255),
    PPT_URL VARCHAR(255),
    Order_No INT NOT NULL,
    FOREIGN KEY (Course_ID) REFERENCES COURSE(Course_ID)
        ON UPDATE CASCADE ON DELETE CASCADE
);

```

5. ENROLLMENT

```

CREATE TABLE ENROLLMENT (
    Enrollment_ID INT PRIMARY KEY AUTO_INCREMENT,
    User_ID INT NOT NULL,
    Course_ID INT NOT NULL,
    Progress DECIMAL(5,2) DEFAULT 0.00,
    Completion_Status VARCHAR(20) DEFAULT 'In Progress',
    UNIQUE (User_ID, Course_ID),
    FOREIGN KEY (User_ID) REFERENCES USER(User_ID)
        ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (Course_ID) REFERENCES COURSE(Course_ID)
        ON UPDATE CASCADE ON DELETE CASCADE
);

```

6. ASSESSMENT

```

CREATE TABLE ASSESSMENT (

```

```
Assessment_ID INT PRIMARY KEY AUTO_INCREMENT,  
Course_ID INT NOT NULL,  
Title VARCHAR(150) NOT NULL,  
Type VARCHAR(50),  
Duration INT COMMENT 'Duration in minutes',  
FOREIGN KEY (Course_ID) REFERENCES COURSE(Course_ID)  
ON UPDATE CASCADE ON DELETE CASCADE  
);
```

7. QUESTION_BANK

```
CREATE TABLE QUESTION_BANK (  
Question_ID INT PRIMARY KEY AUTO_INCREMENT,  
Assessment_ID INT NOT NULL,  
Question TEXT NOT NULL,  
Type VARCHAR(50),  
Difficulty VARCHAR(20),  
Correct_Answer TEXT,  
FOREIGN KEY (Assessment_ID) REFERENCES ASSESSMENT(Assessment_ID)  
ON UPDATE CASCADE ON DELETE CASCADE  
);
```

8. ASSESSMENT_RESULT

```
CREATE TABLE ASSESSMENT_RESULT (  
Result_ID INT PRIMARY KEY AUTO_INCREMENT,  
User_ID INT NOT NULL,  
Assessment_ID INT NOT NULL,  
Score DECIMAL(5,2),  
Marks DECIMAL(5,2),
```

```

Attempt_Date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
Time_Taken INT COMMENT 'Time taken in seconds',
Status VARCHAR(20),
FOREIGN KEY (User_ID) REFERENCES USER(User_ID)
    ON UPDATE CASCADE ON DELETE CASCADE,
FOREIGN KEY (Assessment_ID) REFERENCES ASSESSMENT(Assessment_ID)
    ON UPDATE CASCADE ON DELETE CASCADE
);

```

9. CERTIFICATE

```

CREATE TABLE CERTIFICATE (
    Certificate_ID INT PRIMARY KEY AUTO_INCREMENT,
    User_ID INT NOT NULL,
    Course_ID INT NOT NULL,
    Issue_Date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    Certificate_URL VARCHAR(255),
    QR_Code VARCHAR(255),
    UNIQUE (User_ID, Course_ID),
    FOREIGN KEY (User_ID) REFERENCES USER(User_ID)
        ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (Course_ID) REFERENCES COURSE(Course_ID)
        ON UPDATE CASCADE ON DELETE CASCADE
);

```

10. NOTIFICATION

```

CREATE TABLE NOTIFICATION (
    Notification_ID INT PRIMARY KEY AUTO_INCREMENT,
    User_ID INT NOT NULL,

```

```
Message    VARCHAR(500) NOT NULL,  
Status     VARCHAR(20) DEFAULT 'Unread',  
Sent_Date  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
FOREIGN KEY (User_ID) REFERENCES USER(User_ID)  
    ON UPDATE CASCADE ON DELETE CASCADE  
);
```

11. AI_CHAT

```
CREATE TABLE AI_CHAT (  
    Chat_ID   INT PRIMARY KEY AUTO_INCREMENT,  
    User_ID   INT NOT NULL,  
    Prompt    TEXT NOT NULL,  
    AI_Response TEXT,  
    Timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY (User_ID) REFERENCES USER(User_ID)  
    ON UPDATE CASCADE ON DELETE CASCADE  
);
```

12. CODE_SUBMISSION

```
CREATE TABLE CODE_SUBMISSION (  
    Submission_ID INT PRIMARY KEY AUTO_INCREMENT,  
    User_ID       INT NOT NULL,  
    Question_ID   INT NOT NULL,  
    Language      VARCHAR(50),  
    Source_Code   TEXT,  
    Output        TEXT,  
    Verdict       VARCHAR(50),  
    Score         DECIMAL(5,2),
```

```

FOREIGN KEY (User_ID) REFERENCES USER(User_ID)

ON UPDATE CASCADE ON DELETE CASCADE,

FOREIGN KEY (Question_ID) REFERENCES QUESTION_BANK(Question_ID)

ON UPDATE CASCADE ON DELETE CASCADE

);

```

13. ACHIEVEMENT

```

CREATE TABLE ACHIEVEMENT (

Achievement_ID INT PRIMARY KEY AUTO_INCREMENT,

User_ID INT NOT NULL,

XP INT DEFAULT 0,

Badge VARCHAR(100),

Level INT DEFAULT 1,

Streak INT DEFAULT 0,

FOREIGN KEY (User_ID) REFERENCES USER(User_ID)

ON UPDATE CASCADE ON DELETE CASCADE

);

```

Helpful indexes for common lookups

```

CREATE INDEX idx_user_role ON USER(Role_ID);

CREATE INDEX idx_course_faculty ON COURSE(Faculty_ID);

CREATE INDEX idx_module_course ON LEARNING_MODULE(Course_ID);

CREATE INDEX idx_enrollment_user ON ENROLLMENT(User_ID);

CREATE INDEX idx_enrollment_course ON ENROLLMENT(Course_ID);

CREATE INDEX idx_assessment_course ON ASSESSMENT(Course_ID);

CREATE INDEX idx_question_assessment ON QUESTION_BANK(Assessment_ID);

CREATE INDEX idx_result_user ON ASSESSMENT_RESULT(User_ID);

CREATE INDEX idx_result_assessment ON ASSESSMENT_RESULT(Assessment_ID);

CREATE INDEX idx_certificate_user ON CERTIFICATE(User_ID);

```

```
CREATE INDEX idx_notification_user ON NOTIFICATION(User_ID);
```

```
CREATE INDEX idx_aichat_user ON AI_CHAT(User_ID);
```

```
CREATE INDEX idx_submission_user ON CODE_SUBMISSION(User_ID);
```

```
CREATE INDEX idx_submission_question ON CODE_SUBMISSION(Question_ID);
```

```
CREATE INDEX idx_achievement_user ON ACHIEVEMENT(User_ID);
```